

# BNY MELLON – INTERNSHIP INTERVIEW EXPERIENCE

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The Interview process consisted of 3 rounds:

1. Online Assessment
2. Technical Interview
3. Technical + HR Interview

## Round-1: Online Assessment

The test was conducted in **Hackerrank platform**. There was a **total of 4 questions** and we were given **2hrs 45mins** to solve them.

### 1. Maximizing the Redundancy

You are given  $n$  servers represented as a binary string, where 1 means operational and 0 means not operational. At most one operation can be performed: choose a consecutive segment and flip all values ( $1 \rightarrow 0$ ,  $0 \rightarrow 1$ ).

The task is to find the **maximum redundancy**, defined as the number of unique possible counts of operational servers after performing the operation.

Sample Input: 0011

Sample Output: 4

## 2. Server Clusters

You are given an array `serverProp` representing the properties of  $n$  servers. Two servers at indexes  $i$  and  $j$  are considered connected if the greatest common divisor (GCD) of their properties, `serverProp[i]` and `serverProp[j]`, is greater than 1. Connected servers form clusters in the network.

You need to return an array of integers where the  $i$ -th value represents the size of the cluster to which the  $i$ -th server belongs.

Sample Input:  $n = 3$ , `serverProp = [1, 2, 4]`

Sample Output: `[1, 2, 2]`

## 3. Minimum Cost

There are  $n$  points on the  $x$ -axis  $(1, 2, \dots, n)$ , where each point  $i$  has a cost associated with it `cost[i]`.

Starting from position 0, you can jump forward up to  $k$  steps at a time. Whenever you land on a point, you incur its cost. The goal is to find the **minimum cost** required to reach point  $n$ .

Sample Input:  $n = 5$ , `cost = [4, 3, 9, 3, 1]`,  $k = 2$

Sample Output: 7

## 4. Largest Subset

In a scheduling system, there are  $n$  events represented by two arrays, `start` and `finish`, each of size  $n$ .

- The  $i$ -th event begins at `start[i]` and ends at `finish[i]`.
- A subset of these events is considered **high-priority** if there exists at least one event in the subset that overlaps or intersects with every other event in the subset.

Your task is to determine the **maximum size** of a high-priority subset of events.

Sample Input:  $n = 4$ , `start = [1, 2, 3, 4]`, `finish = [2, 3, 5, 5]`

Sample Output: 3

My performance:

1. 15/15 test cases ✓
2. 14/15 test cases ✓
3. 9/15 test cases ✓
4. 10/15 test cases ✓

## Round-2: Technical Interview

The second round **took place at CUIC**. A total of **26 candidates were shortlisted** from the OA and we were **split across 6 panels** with an interviewer in each.

The interviewer in my panel was very friendly and chill and it helped me initially as I was nervous about the Interview. The interview went for **about 30-40 mins**. After settling in, the interviewer started browsing through my resume and asked me to give **an introduction**. The interviewer then moved on to the technical questions and asked me to choose any of the core concepts I had mentioned in my resume for which I chose **Object Oriented Programming (OOP)** and he started asking some questions about OOP:

- What are the 4 pillars of OOP?
- Explain about any one of them with a real life example.
- Some questions related to relationships in Inheritance.

Then, he moved on to my projects and asked me to explain about the **Real Time Chat Application** project which I had done **using MERN stack**. I explained my part of the project which I worked with. After that he asked me to explain about any one of the projects I had done as per my preference, I explained about the Java project (Spaceship Game) in great detail (use case, functionalities, etc.). Then he tested my knowledge in DSA with a problem:

- I don't remember the exact question but it went like this. How would you design a parking lot by checking if a slot is free for the car to park or not and assign them the free slot.

Initially, I was not really sure whether it was a coding question so I asked the doubts I had and told him my approach and wrote it down in a paper. Even though I made few mistakes in my solution, he corrected me and was trying to test my thinking skills. Then he asked me to explain about any one sorting technique of my choice.

Finally, he asked me if I had any doubts for which I asked **“What kind of work we would be assigned to during the Internship?”** and **“How well did I perform in the interview?”** for which he pointed out my strengths and weaknesses of what he had seen of me in the Interview. I asked some other questions as well.

I was one of the **14 candidates** who got **shortlisted** for the next round.

### **Round-3: Technical + HR Interview**

It took place sometime after the second round got over. I had a long conversation for **about an hour** with the interviewer. The interview started with self-introduction. During the very initial stages, the interviewer pointed out and asked why I had mentioned programming languages in a random order as he assumed it was in the order of my preference and told me most of the people would read it in that way, I was told this in the previous round also so avoid this mistake.

Then, the interviewer asked me various doubts regarding my MERN stack project and asked various hypothetical questions while developing a project and how I would handle the situation. To give a basic idea, the questions were like:

- How would you deal with a customer when you have time constraints on meeting the deadline of a project?
- How would you deal with a situation when your project members are not available?

I told how I would handle the situations and he further complicated the situation of the question and it went on like that. Then, he asked me to name the **aggregate functions** and **some inbuilt functions** from DBMS. Moving on to DSA he asked me 2 questions:

- 4 friends go to a restaurant and pay the bill in a disoriented ratio, how would the split be between each of them and he added further more which I don't really remember much.
- A question related to obtaining a substring from a string.

For the first question, I wrote my approach in a paper logically explaining him and he asked me write the code as well in a language of my preference and I wrote in C++. Although my answer was not perfect, I gave the **logical idea** with coding approach and he helped me further and taught me how to solve the missing part. For the second question, I gave him a **sliding window approach** for which he agreed with.

For major part of the interview, I was asked about my projects and many hypothetical questions. The interviewer also asked me some practical questions related to **Software Engineering Concepts** like SDLC. He also asked me how well do I know Spanish as I had mentioned it in languages in my resume.

Finally he asked me if I had any questions for him, and I questioned with various questions like "How did he handle the situation when he was at my place (candidate)?" and so on. It was a friendly and casual conversation which went on for an extended duration. In the end, he mentioned to me that **he had a great time conversing with me** and the interview ended.

The results were announced after a prolonged wait, and **6 of us were selected** for the Internship ( 4, 2 from MIT, CEG respectively).

### **Tips:**

- Be 100% thorough with all things mentioned in your resume.
- Try not to be anxious and be as confident as possible during the interviews.
- Don't panic and stutter when you don't know the answer for a question, such questions are asked to test your decision making under pressure and rather reply with similar things relevant to the question ("I am not sure on that but I know about this", "I will look into it", etc.) think for sometime and answer I DON'T KNOW only if you have no idea about the question.

- Be honest with the interviewer.
- Respond with creative and thoughtful answers which come to your mind at that moment.
- Present yourself showing your genuine interest for the job opportunity.
- Be curious and ask the interviewer various questions when asked to making the conversation interactive.
- Be thorough with the core concepts you have learned up until at the time of interview **especially DSA, DBMS, OOP**.
- Be prepared on what you tell for self-introduction, this is crucial as the interviewer gets keen and curious about the strengths, hobbies and other things you mention.
- Create a habit of practicing DSA problems regularly in coding platforms like LeetCode to become more familiar with DSA concepts.

Don't get vexed if you are not hired, try your best and the good luck, results will eventually find you in the future.

Best of luck! Please do feel free to reach out to me if you have any further doubts,

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